

# INTRODUCTION

## Defining Quality

When the expression “quality” is used, we usually think in terms of an excellent product or service that fulfills or exceeds our expectations. These expectations are based on the intended use and the selling price. For example, a customer expects a different performance from a plain steel washer than from a chrome-plated steel washer because they are a different grade. When a product surpasses our expectations, we consider that quality. Thus, it is somewhat of an intangible based on perception. Quality can be quantified as follows:

$$Q = \frac{P}{E}$$

where Q= quality

P = performance

E = expectations

If Q is greater than 1.0, then the customer has a good feeling about the product or service. Of course, the determination of P and E will most likely be based on perception with the organization determining performance and the customer determining expectations.

More definitive definition of quality is given in ISO 9000: 2000. It is defined as the degree to which a set of inherent characteristics fulfills requirements. Degree means that quality can be used with adjectives such as poor, good, and excellent. Inherent is defined as existing in something, especially as a permanent characteristic. Characteristics can be quantitative or qualitative. Requirement is a need or expectation that is stated; generally implied by the organization, its customers, and other interested parties; or obligatory.

❑ Fitness for purpose or use – Juran

❑ Quality is conformance to requirements -Crosby

❑ Quality shall be aimed at the needs of the consumer present and future -Deming

❑ The totality of the features and characteristics of a product or service that bears on its ability to satisfy stated or implied needs - ISO (International organization of standardization)

## Concept of quality

- ✓ Well-designed product with functional perfection first time right
- ✓ Provide satisfaction beyond customer's expectation.
- ✓ Excellence in service
- ✓ Absolute empathy with customer
- ✓ Quality consists of freedom from deficiencies

**The basic building block on which fitness for use is built is the quality characteristic, a) Technological, b) Psychological, c) Time oriented, d) Contractual and e) Ethical.**

## Quality Objectives

Quality objectives should be SMART

- A. S for Specific
- B. M for Measurable
- C. A for Attainable
- D. R for Return
- E. T for Timely

## Quality of products

There are many various features of product quality:

### **a) Functionality**

This refers to the main characteristics of a product and a service. It is defined as a set of alternatives that are related to the existence of a set of functions and their specified properties. All functions have specific properties that explain the quality of the product and its services. The functions are also those that justify the declared needs for the manufacture of qualitative products.

### **b) Reliability**

Reliability is also the main feature of a quality product. Measurement of product reliability is done by finding the 'mean time between failures' (MTBF). It is an indicator of durability of products.

### **c) Usability**

Usability is also a hallmark of product quality. This shows the client's ability to use the product. It is expected that the client can use any type of product without the help of experts.

### **d) Maintainability**

This refers to the ease of a product being able to be kept in its original state without creating a problem. The ability to keep a product for a while makes it expensive, but you may finally use a product for a long time. It may be defective during use and must be repaired to maintain its original quality at the lowest cost. It is measured as 'Average Repair Time (MTTR).'

### **e) Efficiency**

Efficiency is the relationship between production and input. It means what we do after consuming sources of inputs. Therefore, a quality product must be measurable by its effectiveness.

### **f) Portability**

The quality of the product also has the particularity of portability. It is defined as a set of attributes that affect the ability of the software to transfer from one environment to another. It simply transfers different attributes from one place to another. The environment can be hardware, software or an organization.

## **Dimensions of quality and their measure**

Garvin (1984) provides discussion of eight critical dimensions of product quality. The summarized key points concerning these dimensions of quality are provided below.

### **a) Performance**

This is the fundamental dimension of product quality. Discuss the performance evaluation of the product/service based on specific features, characteristics, and learn how it works from the customer's viewpoint.

### **b) Reliability**

This is about the quality of a product based on its reliability, the probability of not having faulty components in products such as cars or airplanes during their service in a given period. It must be reliable, because its reliability threshold is lower, plus there are the possibilities of repair or replacement. This is why it affects the quality of the product.

### **c) Durability**

Durability is the effective life or longevity of the product before it is declared unusable. Repair is not possible after this phase of life. This means that when the product is not sustainable, it completes its useful life.

### **d) Serviceability**

This is the ability to provide a service to customers. The customer's opinion of quality is also influenced by the speed and economic profitability of a routine maintenance activity. Show how much your company is serious about your product services.

### **e) Aesthetics**

This concerns the visual appearance or the appearance of the product, often taking into account factors such as model, color, style, packaging, tactile characteristics, and other sensory characteristics.

### **f) Features**

The tendency of customers is usually to go for a product with more and more value-added features. This can go beyond the basic criteria to enter the market. Characteristics are also the fundamental dimension of product quality. A feature can also be defined as an additional or secondary attached feature, and complements the main function of a product.

### **g) Perceived quality**

This signifies the reaction of a customer after the use of the product and/or service. The product must have a perceived quality that enhances customer satisfaction. This dimension is directly affected by any product failure that is highly visible to the public or by the way the service is delivered to the customer when a product quality issue is resolved. Customer loyalty and brand loyalty are closely tied to perceived quality.

### **h) Conformance to Standards**

Compliance with standards is also a fundamental dimension of quality. Conformance to quality provides the standard for value.

## **Improving productivity through quality**

Productivity and quality management are key factors in competitiveness and always preoccupied the productive sectors, especially in open economies. Quality is the fundamental factor by which improvisation intervenes in productivity. Targeted efforts improve productivity, and quality management of an organization often results in greater internal and external satisfaction because costs and reliability are met. Here are some factors of quality through which productivity can improve:

### **a) Analyze process**

The whole process needs to be analyzed where the focus needs to be on the processes, and not on those who perform those processes. The policy to standardize procedures of the company helps in maximizing the efficiency of the company. Follow the standard value for any type of product. When appropriate training is provided to all the staff, they can produce high quality products in a short period of time.

### **b) Align business processes with other companies in the sector**

Find out how other competing companies are organizing and conducting business in a way that allows for the integration of the business process and, as a result, improvements. It needs to be well-organized and have alternatives planned for any type of operation. All information is important because it can save time and money. This exercise also provides good ideas for improvement.

### **c) Develop performance measures**

To improve productivity, managers should appreciate the quality of developed performance parameters. Problem identification, prediction of outcomes of the future, and productivity gains measurement with key sectoral performance indicators should use current processes as the point of reference.

#### **d) Build quality tests in processes**

In each organization, to improve productivity through quality, quality testing must be integrated into the process. Experience is gained during the process and not just at the end, so it provides options and alternatives for better results. This does not create a situation when it is more expensive to repair. Automated tests can be performed which takes less time to provide results in processes that are easy to explain, modify, correct, approve, or reject.

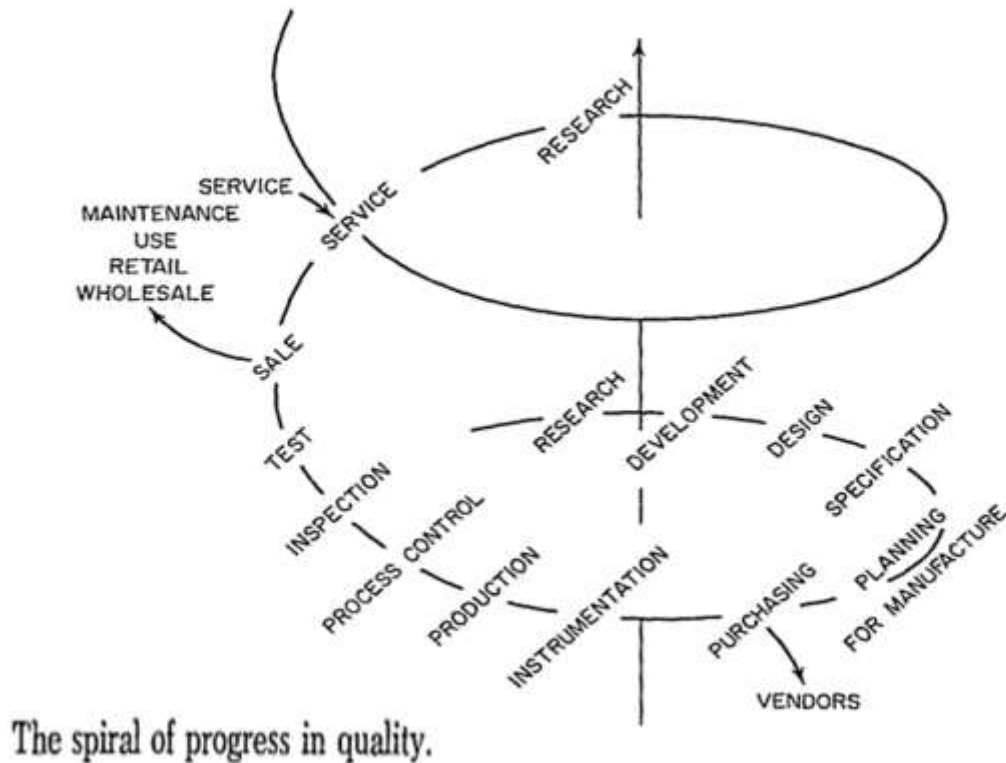
#### **e) Use business strategies to improve productivity and quality management**

To improve productivity through quality, always develop business strategies and use them. It helps in identifying the root cause of problems and intervenes to improve or optimize processes. In quality management, production must be a control so that defects are corrected immediately before they affect their final product or result.

#### **f) Pay attention to feedback**

Feedback collection from all the stakeholders likes employees, customers, suppliers, vendors, and business partners for their opinion on products or services. Provides information on whether the selected route is correct or not. Organizations use customer feedback to improve current products and influence the design of new products.

## Quality Function



## Quality Guru's Theory

### 1. Deming's 14 points

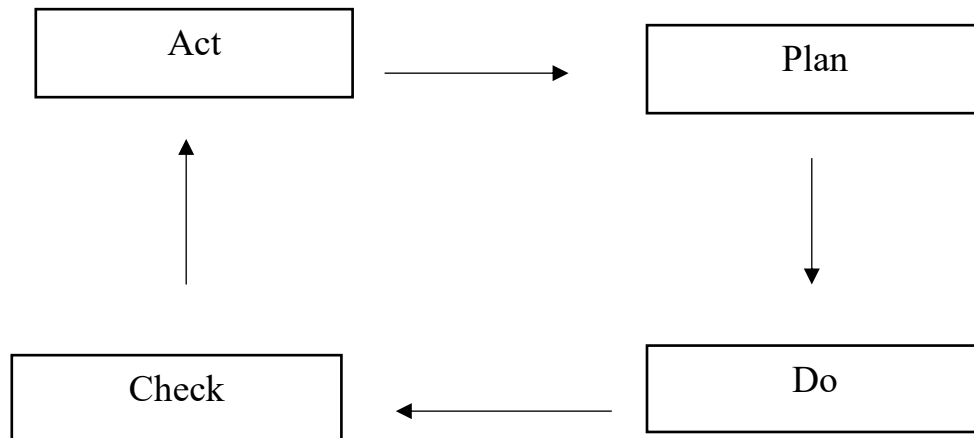
1. First, create reliability of purpose toward enhancement of product and service, with the aim of becoming competitive, remaining in business, and providing jobs.
2. Adopt the new philosophy. Western management must be stimulated by the challenge, must learn their responsibilities, and take on leadership for change.
3. End dependence on mass inspection; build quality into the product from the beginning.
4. End the practice of granting business on the basis of price tag alone. Instead, minimize total cost. Select based on a long-term connection of loyalty and trust.

5. Improve continuously and forever the system of production and service to improve quality and reduce waste.
6. Introduce training and retraining.
7. Institute leadership. The aim of observation should be to lead and aid people in discharging their duties better.
8. Drive out fear so that everyone may work effectively for the company.
9. Remove the hurdles step-by-step between various departments like research, design, sales, and production to help them work cohesively, more as a group, to predict and solve problems in production.
10. Eliminate slogans, exhortations, and targets for the workforce, as they do not necessarily achieve their aims.
11. Eliminate numerical quotas in order to take account of quality and methods, rather than just numbers.
12. Remove barriers to pride of workmanship.
13. Introduce a dynamic programme of education and re-training for both the management and the workforce.
14. Management and workforce must work together.

### **The Deming cycle**

Shewhart invented the concept of Plan-Do-Check-Act (PDCA cycle), which was later on propagated by Dr. Deming. Their cyclic approach for planning and testing purposes is to make a full-scale implementation of improvement and finally formalizing it. This cycle provides the complete information of evaluation and basic ideas. An improvement idea needs testing on a smaller scale before it can be utilized for full-scale implementation. By doing small-scale testing, validation of its benefit is possible before large-scale implementation is undertaken. Furthermore, resistance from the employees is also reduced owing to more time being available for them to accept and adjust to the newer methods. This way it is more likely that employees will support such initiatives.

The Deming PDCA Cycle provides opportunities for continuous evaluation and improvement. The steps in the Deming PDCA or PDSA Cycle as shown here.



1. First step is to make a plan for an improvement causing change or test (P).
2. Second step is to do it (D). To conduct small-scale testing of the planned improvement change.
3. Third step is to check it (C). To check the effects arising from the conducted test. It is also called ‘study’ sometimes (S).
4. Next and last step is to act on learning from the first step (A).
5. Keep repeating Step 1, with knowledge generated from previous steps

## 2. Philip Crosby philosophy

Philip Crosby is a noted quality professional, author, and consultant who is widely known for promoting the concept of “zero defects” and for attempting to define quality from the viewpoint of conformance to requirements.

### The Fourteen Steps to Quality Improvement

#### 1. Management Commitment

Make it clear that management is committed to quality.

## 2. Quality Improvement Teams

Form Quality Improvement Teams with senior representatives from each department.

## 3. Measure Processes

Measure processes to determine where current and potential quality problems lie.

## 4. Cost of Quality

Evaluate the cost of quality and explain its use as a management tool.

## 5. Quality Awareness

Raise the quality awareness and personal concern of all employees.

## 6. Correct Problems

Take actions to correct problems identified through previous steps.

## 7. Monitor Progress

Establish progress monitoring for the improvement process.

## 8. Train Supervisors

Train supervisors to actively carry out their part of the quality improvement program.

## 9. Zero Defects Day

Hold a Zero Defects Day to reaffirm management commitment.

## 10. Establish Improvement Goals

Encourage individuals to establish improvement goals for themselves and their group.

## 11. Remove Fear

Encourage employees to tell management about obstacles to improving quality.

## 12. Recognize

Recognize and appreciate those who participate.

## 13. Quality Councils

Establish Quality Councils to communicate on a regular basis.

## 14. Repeat the Cycle

Do it all over again to emphasize that the quality improvement process never ends.

### **3. The Juran Trilogy**

The Juran Trilogy was developed by Dr. Joseph Juran. The Juran Trilogy is an improvement cycle that is meant to reduce the cost of poor quality by planning quality into the product/process.

#### **The Juran Trilogy**

##### **1. Quality Planning**

In the planning stage, it is critical to define who your customers are and find out their needs (the “voice of the customer”). After you know what your customers need, you’re able to define the requirements for your product/process/service/system, etc., and develop it. Additionally, any plans that might need to be transferred to operators or other key stakeholders should be done during the planning phase. Planning activities should be done with a multidisciplinary team, with all key stakeholders represented.

##### **2. Quality Control**

During the control phase, determine what you need to measure (what data do you need to know if your process is working?), and set a goal for your performance. Get feedback by measuring actual performance, and act on the gap between your performance and your goal. In Statistical Process Control (SPC), there are several tools that could be used in the “control” phase of the Juran Trilogy: Pareto Analysis, flow diagrams, fishbone diagram, and control charts, to name a few.

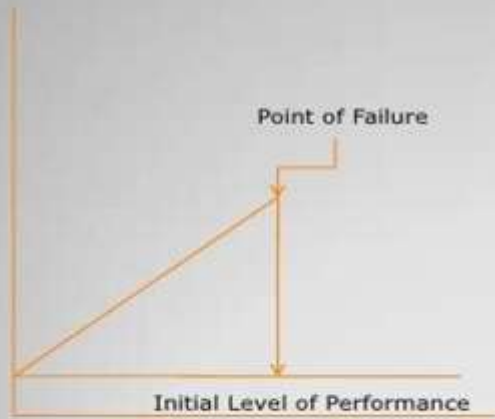


### **3. Quality Improvement**

There are four different “strategies” to improvement that could be applied during this phase:

- Repair: Reactive; fix what’s broken.
- Refinement: Proactive; continually improve a process that isn’t broken (like the continual pursuit of perfection in Lean!)
- Renovation: Improvement through innovation or technological advancement
- Reinvention: Most demanding approach; start over with a clean slate.

## Continuous v/s Continual Improvement

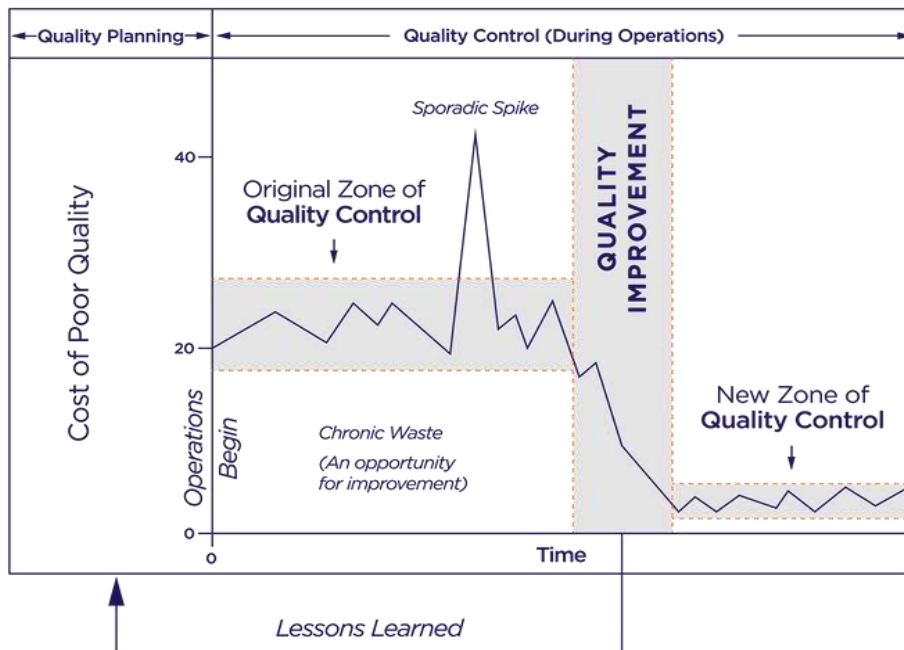


Continuous Improvement



Continual Improvement

## THE JURAN TRILOGY®



## Quality Control

Measurement of the actual performance compare with standard if there is any variation act on the variation or difference

### Act ▪ Preventive action ▪ Correction ▪ Corrective action

**Defect:** No fulfilment of a requirement related to specific use

**Preventive action:** Action to eliminate the cause of a potential non conformity

**Correction:** Action to eliminate a detected non conformity

**Corrective action:** Action to eliminate the root cause of a detected non conformity

## Impact of poor quality



## Cost of Quality

Quality cost, also known as the Cost of Quality (COQ), refers to the total cost associated with achieving and maintaining a desired level of quality in a product or service. It encompasses all costs related to preventing defects, appraising quality, and addressing both internal and external failures.

The cost of quality is generally classified into four categories:

1. External failure cost.
2. Internal failure cost.
3. Inspection (appraisal) cost.
4. Prevention cost

## Cost of conformance

Cost of conformance is defined as the element of cost to avoid poor quality of products. It consists of costs associated with quality-assuring activities, like process standardization, training programs, and costs associated with quality controlling activities, like reviewing, auditing, inspecting, and testing.

Cost of conformance is then divided into two types

- 1) Prevention cost.
- 2) Appraisal cost.

The preventive costs are the costs due to enhancing the quality in production and servicing so that any fluctuation in process will be accepted either on upper or lower range of control charts. These costs generally consist of workers' guidance costs and equipping the workplace with advanced machinery and technologies. These also involve activities in the documentation process. These are the activities that costs are associated with and incurred:

- Collecting information on consumers' needs by proper surveying.
- Transfer of consumers' needs into the design and documentation stage.
- Checking the authentication of suppliers.
- Proper training of employees to enhance their capabilities.
- Quality administration.

Appraising cost is another element in this category. This cost includes standardization of the processes and activities so that a high level of quality in products and processes can be achieved. These costs are further incurred by the following activities or processes as elaborated below.

- Testing and inspecting the data.
- Inspecting the facilities provided by the suppliers.
- Evaluating the quality during packaging and dispatching.
- Performing different types of non-destructive testing.
- Field trials of new products.

## **Failure cost**

This cost has a great and immediate impact on any organization's performance. Whenever a breakdown occurs, there will always be lost profit. This cost may include following elements

- Lost profit.
- The cost of the repair.
- The constant and unpredictable operational wages exhausted during the downtime.

Usually, these costs are huge for any organization per year. These costs are normally recognized by top management even after bankruptcy. Failure costs are mainly incurred as during failure, whether it is of machinery or manpower, no production takes place, though inventory carrying costs and other maintenance costs or repairing costs are incurred continuously. Thus, fixed costs are continuously increasing, but there is no production taking place. Sometimes these costs are difficult to avoid for department managers and other top management.

There are two types of elements into which failure cost can be classified namely:

1. Internal failure cost.
2. External failure cost.

Total external and internal cost includes the following elements:

1. Labor: Both direct and indirect.
2. Product waste: This includes product repair and rework costs.
3. Services: This includes servicing of failed machinery and equipment cost.
4. Cost incurred due to inadequate materials supplied by suppliers.
5. Some additional costs of miscellaneous failures.

## **Internal Failure cost**

This cost occurs with non-conforming products and services namely:

- a. Product scrapping costs: This cost is associated with non-usable products.
- b. Product or machinery repair costs.
- c. Work environment reconditioning costs.

Internal failure costs consist of the following elements of non-conformance costs:

- a. Failure cost during the designing phase of the products: This cost is mainly incurred due to inadequate design, reworking on design changes, and scrap produced due to design changes of products.
- b. Failure cost due to purchasing: This cost is mainly incurred due to rejection of outsourced equipment and other raw materials, replacement of out sourced facilities, suppliers' action correction, etc.
- c. Operational failure cost: This cost includes a major portion of internal failure cost due to operationally unsuitable products or services, resource reviewing, product scrapping costs, internally active labor costs, etc.

## **External failure cost**

This cost is mainly incurred after products or services have been delivered to consumers. This cost may include the following elements:

- a. Eliminating customers' complaints, and providing home services.
- b. Back-ordering products for the consumers.
- c. Redesigning costs due to faulty design steps.
- d. Warranty costs paid by the organization for repairing, cleaning, and poor working condition of the product.
- e. Costs paid for damage and destruction to consumers and workplaces by products delivered by the organization.
- f. Penalties imposed by government regulatory bodies on the organization due to its poor performance.
- g. Cost paid to the dissatisfied consumers in the form of rewards or cashback.
- h. Costs incurred by the organization to prevent the loss due to fall in sales.

## Total quality management

TQM can be summarized as a management system for a customer-focused organization that engages all employees in continual improvement of the organization. It is an integrative system that uses strategy, data, and effective communications to integrate the quality discipline into the processes, products, services, and culture of the organization. The methods for implementing this approach are found in the teachings of quality leaders such as Philip B. Crosby, W. Edwards Deming, Armand V. Feigenbaum, Kaoru Ishikawa, and Joseph M. Juran.

## Key principles of TQM

**1. Customer focused:** The primary goal of TQM is to meet or exceed customer expectations. Understanding customer needs and providing products or services that deliver value is a central tenet.

**2. Employee involvement:** All employees, regardless of their level or role, should actively participate in quality improvement efforts. Training, empowerment, and fostering a culture of accountability are essential.

**3. Process approach:** TQM emphasizes understanding, managing, and improving processes to achieve desired results. It focuses on efficiency and effectiveness in the operations of an organization.

**4. Integrated system:** TQM integrates all organizational systems and processes to work toward common quality objectives. Goals are aligned across teams to ensure coherence and collaboration.

**5. Strategic and systematic approach:** The quality management system is aligned with the organization's strategic goals and is consistent with long-term objectives.

**6. Continual improvement:** The organization strives for ongoing improvements in processes, products, and services. Incremental enhancements and innovative changes contribute to overall quality.

**7. Fact-based decision making:** The organization makes decisions based on accurate data and objective analysis rather than assumptions. Statistical tools and quality management methods can aid in this process.

**8. Communications:** Communication is the glue that binds the principles of TQM together, ensuring seamless implementation and promoting a culture of quality throughout the organization. It fosters a shared vision, encourages employee engagement, promotes transparency, and enhances collaboration.

## **Essential requirements for successful implementation of TQM**

### **a) Commitment**

Quality improvement (in all aspects) must be the work of all members of the organisation. A clear commitment on the part of management must be provided, removing barriers to continuous quality improvement and the steps necessary to create an environment conducive to attitudinal change. Training and support in this regard should be expanded.

### **b) Culture**

Culture is a step for successful implantation of TQM. Therefore, before changes take effect in the culture and attitude of the organisation, suitable training and awareness is necessary.

### **c) Continual Improvement**

Improvement needs to be understood as not only a one-time affair. Rather, it is a process that is continuous in nature. It is basically a need of the customer, as well as of the service provider.

### **d) Customer Focus**

A customer-oriented approach ensures the intention of the organization is to attain perfection with no defects and giving comprehensive satisfaction to the end-user. The users could be internal to the company or external.

### **e) Control**

A process control exercise through monitoring and regular checks ensures no deviation from the intended course of implementation.

## **Benefits of TQM**

TQM offers the following benefits:

- **Less product defects.** An objective of TQM is to create products and services correctly the first time. This means that products ship with fewer defects, reducing product recalls, future customer support overhead and product fixes.
- **Satisfied customers.** High-quality products that meet customers' needs result in higher customer satisfaction. High customer satisfaction, in turn, can lead to increased market share, revenue growth via upselling and word-of-mouth marketing initiated by customers.
- **Lower costs.** As a result of less product defects, companies save money on customer support, product replacements, field service and creating product fixes. The cost savings flow to the bottom line, creating higher profit margins.
- **Well-defined cultural values.** Organizations that practice TQM develop and nurture core values around quality management and continuous improvement. The TQM mindset pervades across all aspects of an organization, from hiring to internal processes to product development.

## Disadvantages of TQM

The downsides of TQM include the following:

- **Planning and resources.** TQM needs a significant amount of planning and resources over time to be properly allocated to the change.
- **Companywide commitment.** Continuous improvement in TQM means the organizational culture must focus on improving processes. All management levels must be supportive.
- **Added costs.** TQM might add training, infrastructure and team development costs.
- **Time.** It might take years for an organization to fully show intended results.
- **Partial efforts.** Because of the effort involved in implementing TQM, a partial move toward it can result in failure.

## Implementation steps of TQM

The relevant prerequisites relate to the history of the organization, its current needs, the events that led to the management of total quality, and the quality of working life of employees. If the current reality does not include important preconditions, the implementation of TQM should be delayed until the organization is in a favorable state for its success. If an organization has a proven track record of environmental awareness and has been able to effectively change the way it operates, when necessary, TQM will be easier to implement. If an organization has always been unreceptive and lacks the capacity to improve its operating systems, employees will be unhappy and qualified change agents will be lacking. If this condition prevails, a comprehensive management and leadership development program can be established. A management audit is a good evaluation tool to identify the current levels of operation of the organization and the areas that require changes. An organization must be fundamentally healthy before starting TQM.



Implementation steps of TQM.

## **Obstacles to implementing TQM**

Organizations have a number of difficulties in implementing TQM. A few common obstacles are:

### **a) Lack of management commitment**

When management talks about TQM, but its actions fail, it will eventually not be able to meet expectations. The result is distrust and the difficulty of launching another attempt. For the implementation to be successful, the administration must clearly communicate the reason for adopting TQM.

### **b) Inability to change organizational culture**

Changing the culture of an organization is very difficult and takes time. The fear of change must be addressed, conflicts between the social partners and companies must be resolved, and the focus of the organizations must no longer be maintained. Without this, no-one will have confidence in the organization.

### **c) Improper planning**

All parts of the organization must participate in the development of the implementation plan and the changes as the plan evolves.

### **d) Lack of continuous training and education**

TQM raises the problem of adequate education and training. Training and education are most effective when senior management leads the training on the principles of TQM.

### **e) Ineffective measurement techniques and lack of access to data and results**

TQM depends on data-driven decision-making. Access to data and rapid recovery are necessary for an efficient process. A process to create and maintain a TQM environment must be accurate, fast, and reliable.

### **f) Paying inadequate attention to internal and external customers**

TQM does not pay enough attention to internal and external customers. It must understand the changing needs and expectations of customers. To avoid this obstacle, it is necessary to find the right way to get directly to the client.